

# Enumerative combinatorics

For a finite set  $A$ , we denote:

1)  $|A|$  - the number of elements (the cardinality of  $A$ )

2)  $A \times B = \{(a, b) : a \in A, b \in B\}$  - Cartesian product

3)  $A \cup B$  - union

4)  $A \cap B$  - intersection

5)  $A \setminus B$  - difference

## 1. Counting by bijections

Def. 1) Let  $f: A \rightarrow B$  be a function:

-  $f$  is injective if different elements have different images:  $x \neq y \Rightarrow f(x) \neq f(y)$

-  $f$  is surjective if every element of  $B$  is the image of an element of  $A$ .

-  $f$  is bijective if it is injective and surjective, that is, if every element of  $B$  is the image of exactly one element of  $A$ .

Obs. A function  $f: A \rightarrow B$  is bijective  $\Leftrightarrow f$  is invertible, that is, there is  $g: B \rightarrow A$  with  $g(f(x)) = x, \forall x \in A$  and  $f(g(y)) = y, \forall y \in B$ .

Def. 2) A set  $A$  has  $n$  elements,  $|A|=n$ , if there is a bijection  $f: \{1, 2, \dots, n\} \rightarrow A$ . The empty set has 0 elements.

Th. 3) (The bijection principle) Let  $A$  and  $B$  be two finite sets:

a)  $|A|=|B| \Leftrightarrow$  there is a bijection  $f: A \rightarrow B$

b)  $|A| \leq |B| \Leftrightarrow$  there is an injective function

$f: A \rightarrow B$

Method. When a set is hard to count directly, we look for a bijection between it and a set that is easy to count. To show that a correspondance is bijective, we also describe the inverse procedure ("the way back").

ex.: 3, 10, 17, 24, ..., 2026  $\div 7$

7, 14, 21, 28, ..., 2030  $\div 7$

1, 2, 3, 4, ..., 290  $\Rightarrow 290$  nr

$$100 \cdot 7 - 7 = 700 - 7 = 696$$

$$f(a) = \frac{(a+7)}{7} \quad f: \{3, 10, \dots, 2026\} \rightarrow \{1, 2, \dots, 290\}$$

Obs. (Watch out for  $+1$ ). An arithmetic progression  $a, a+n, a+2n, \dots, b$  has  $\frac{b-a}{n} + 1$  terms.

$$\begin{array}{l} \text{ex.: } 3^0 = 1 \\ 3^1 = 3 \\ 3^0 + 3^1 = 4 \end{array}$$

$$\begin{array}{l} 2^0 = 1 \\ 2^1 = 2 \\ 2^0 + 2^1 = 3 \end{array}$$

$$\begin{aligned} 3^2 &= 9 \\ 3^0 + 3^2 &= 10 \\ 3^1 + 3^2 &= 12 \\ 3^0 + 3^1 + 3^2 &= 13 \end{aligned}$$

$$\begin{aligned} 2^2 &= 4 \\ 2^0 + 2^2 &= 5 \\ 2^1 + 2^2 &= 6 \\ 2^0 + 2^1 + 2^2 &= 7 \end{aligned}$$

$$202600_{(10)} = 161021220201_{(3)}$$

| 202600 | 3 |
|--------|---|
| 67533  | 1 |
| 22511  | 0 |
| 7503   | 2 |
| 2501   | 0 |
| 833    | 2 |
| 277    | 2 |
| 92     | 1 |
| 36     | 2 |
| 10     | 0 |
| 3      | 1 |
| 1      | 0 |
| 0      | 1 |

$$\Rightarrow 101011111111_{(3)} =$$

$$= 200110_{(10)}$$

$$= 2^{11} + 2^9 + 2^7 + 2^6 + \dots + 2^0$$

$$= 2815 \text{ number}$$

$$2026_{(10)} = 1111101010_{(2)} = 2812_{(10)}$$

## 2. The product rule (the rule of "and")

ex.  $A \rightarrow B$  6 routes  $\Rightarrow 6 \cdot 5 = 30$  possibilities  
 $B \rightarrow C$  5 routes

Th. 1) (The product rule) For any finite sets  $X_1,$

$X_2, \dots, X_n$ :

$$|X_1 \times X_2 \times \dots \times X_n| = |X_1| \cdot |X_2| \cdot \dots \cdot |X_n|$$

2) (The product rule, in words) Each object of a set  $M$  is built by a sequence of  $n$  choices, made one after the other, so that:

(i) at step  $i$  there are exactly  $m_i$  options, whatever the choices at the previous step.

(ii) each object of  $M$  is obtained from exactly one sequence of choices.

$$\text{Then } |M| = m_1 \cdot m_2 \cdot \dots \cdot m_n$$

$$\text{ex.: } \underline{9} \underline{9} \underline{8} = 9 \cdot 9 \cdot 8 = 648 \text{ de numere}$$

### 3. Double counting and the division rule:

ex.  $A, B, C, D, E, F, G, H$

$$\begin{array}{l} 1 \text{ player} \rightarrow 7 \text{ pos.} \\ 2 \text{ player} \rightarrow 5 \text{ pos.} \\ 3 \text{ player} \rightarrow 3 \text{ pos.} \\ 1 \text{ player} \rightarrow 1 \text{ pos.} \end{array} \quad \left| \Rightarrow 7 \cdot 5 \cdot 3 \cdot 1 = 105 \right.$$

$A, B, C, D$

$$\begin{array}{l} 1 \text{ player} \rightarrow 3 \text{ pos.} \\ 2 \text{ player} \rightarrow 1 \text{ pos.} \end{array} \quad \left| \right.$$

$$\Rightarrow 3 \cdot 1 = 3$$

Ob. Where there is no order, we impose one.

Th. 1) (The division rule) If  $f: A \rightarrow B$  and every element of  $B$  is the image of exactly  $k \geq 1$  elements of  $A$ , then  $|A| = k|B| \Rightarrow |B| = \frac{|A|}{k}$ .

Obs. 1) A set with  $n$  elements has  $2^n$  subsets

2)  $2m$  people can be split into pairs in:

$$\frac{(2m)!}{2^m \cdot m!} = (2m-1)!! = (2m-1)(2m-3) \dots 3 \cdot 1 \text{ ways}$$